

VS Code Reference

Optional background — not required for HW1

What is VS Code?

Think of VS Code as a **well-equipped workshop**:

- **Tools** – Built-in coding features
- **Tool Storage** – Extensions for specialized tasks
- **Workbench** – Clean, organized workspace
- **Magnifying Glass** – Debugging & analysis
- **Reference Books** – Integrated documentation

Why Choose VS Code?

- **Free & Open Source** – No licensing fees, community-driven
- **Lightweight but Powerful** – Fast startup, rich features
- **Universal Language Support** – Works with 50+ languages
- **Extensible** – 40,000+ extensions to customize your workflow
- **Integrated Development** – Git, terminal, debugger built-in
- **Cross-Platform** – Same experience on Windows, macOS, and Linux

What Can VS Code Do?

- **Smart Editing** – Syntax highlighting, IntelliSense, snippets
- **Code Navigation** – Go to definition, find references, symbol search
- **Debugging** – Breakpoints, step-through, variable inspection
- **Version Control** – Git integration, visual diffs, commit management
- **Project Management** – Multi-folder workspaces, file explorer, integrated terminal

Command Palette & IntelliSense

- **Command Palette:** `Ctrl+Shift+P` / `Cmd+Shift+P` — one search bar
for all VS Code features, fuzzy search
- **IntelliSense**, just like predictive text on your phone:

```
pr → print()
```

```
console. → log() | error() | warn()
```

Example: Python Project

1. Open Folder: `my_first_project`

2. Create File: `hello.py`

3. Write Code:

```
print("Hello, VS Code!")  
name = input("Your name: ")  
print(f"Nice to meet you, {name}!")
```

4. Run: Right-click → *Run Python File in Terminal*

Debugging Basics

- **Breakpoints** – Stop at specific lines
- **Step Through** – F10 (over), F11 (into), F5 (continue)
- **Inspect Variables** – Hover to see values, use Watch panel

You'll need language-specific extensions (like PHP Debug) installed to debug that language.

Customizing VS Code

Themes: File → Preferences → Color Theme

Keyboard Shortcuts: File → Preferences → Keyboard Shortcuts

Settings: File → Preferences → Settings (Ctrl+,)

- **Project-specific:** `.vscode/settings.json` in your folder
- **Global:**
 - Mac: `~/Library/Application Support/Code/User`
 - Windows: `%APPDATA%\Code\User`

Git Integration & Search

- **Stage Changes:** Click `+` next to files
- **Commit:** Add a message, click ✓
- **Push/Pull:** Sync with GitHub or other repos
- **Find in File:** `Ctrl+F` / `Cmd+F`
- **Find in All Files:** `Ctrl+Shift+F` / `Cmd+Shift+F`
- **Quick Open File:** `Ctrl+P` / `Cmd+P`

Essential Shortcuts

Action	Windows/Linux	macOS
Command Palette	Ctrl+Shift+P	Cmd+Shift+P
Quick Open	Ctrl+P	Cmd+P
Toggle Terminal	Ctrl+`	Cmd+`
Comment Code	Ctrl+/*	Cmd+/*
Format Document	Shift+Alt+F	Shift+Opt+F

VS Code Extensions

Supercharge Your Editor

Extensions: Add New Powers

Need	Extension
Python Dev	Python, Pylance
Web Dev	Live Server
Git Tools	GitLens
Formatting	Prettier
Themes	Material Theme

How to Install

- **Marketplace:** Extensions icon → Search → Install
- **Command Palette:** Ctrl+Shift+P → "Extensions: Install"
- **CLI:**

```
code --install-extension ms-python.python  
code --install-extension esbenp.prettier-vscode
```

Key Extensions by Category

Python

Python · Pylance · Jupyter · Black Formatter

Web Development

Live Server · HTML CSS Support · ESLint · Emmet · Auto
Rename Tag

Git & Version Control

GitLens · Git Graph · GitHub Pull Requests

Debugging & Testing

Jest · Test Explorer UI

Code Quality & Formatting

Prettier · SonarLint

Productivity

**GitHub Copilot · Bracket Pair Colorizer · Path Intellisense ·
TODO Highlight**

Remote Development

Remote - SSH · Remote - WSL · Dev Containers

Database & API

SQLTools · MongoDB for VS Code · PostgreSQL · REST Client

Documentation

Markdown All in One (Markdown editing helpers — not a Marp slide

renderer) · **Draw.io Integration · Better Comments**

Team Collaboration

Live Share · CodeTour · Settings Sync



Top 10 Extensions for Software Engineers

1. **Python** – Core Python support
2. **Pylance** – Better IntelliSense
3. **Live Server** – Instant web preview
4. **Prettier** – Code formatter
5. **ESLint** – JavaScript/TypeScript linting
6. **GitLens** – Git history & blame (or Git Graph / Git History)
7. **Bracket Pair Colorizer** – Easy code navigation
8. **Path Intellisense** – File path autocomplete
9. **Markdown All in One** – Better docs
10. **Live Share** – Real-time collaboration